

Mathematics Stage 3 -- Year B -- Unit 37

Multiplicative Relations & Measurement

KLA: Mathematics | Stage 3 | Year 6 | Term: _____ | Duration: 2-3 weeks | Teacher: _____ | Year: _____

Syllabus outcomes

Code	Outcome descriptor
MAO-WM-01	Develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly.
MA3-MR-01	Selects and applies appropriate strategies to solve multiplication and division problems.
MA3-GM-02	Selects and uses the appropriate unit and device to measure lengths and distances including perimeters.
MA3-2DS-02	Selects and uses the appropriate unit to calculate areas, including areas of rectangles.
MA3-2DS-03	Combines, splits and rearranges shapes to determine the area of parallelograms and triangles.
MA3-3DS-02	Selects and uses the appropriate unit to estimate, measure and calculate volumes and capacities.

Syllabus content points (addressed through SOLO tracker)

All content points below are covered by students working independently through the SOLO tracker. Mini masterclasses target specific outcomes based on live tracker data.

Red -- Foundation	Yellow -- On-level	Green -- Extension
Multiplicative relations A, 2D spatial structure A, geometric measure A Use the term product; multiply numbers; find factors of a whole number (MR-A). Determine whether a number is prime, composite or neither (MR-A). Use the area model to multiply 2-digit by 1-digit numbers (MR-A). Calculate areas of rectangles in cm ² , m ² and km ² (2DS-A). Recognise same-perimeter rectangles can have different areas (GM-A / 2DS-A). Find the volume of a rectangular prism by layers; calculate it in cm ³ (3DS-B).	2D & 3D spatial structure B, multiplicative relations B (applying) Transform a parallelogram into a rectangle by dissection; calculate its area (2DS-B). Recognise a triangle as half a parallelogram; calculate the area of a triangle (2DS-B). Explain a cubic metre and calculate volumes of rectangular prisms in m ³ (3DS-B). Select efficient strategies to multiply up to 4-digit numbers by 1- and 2-digit numbers (MR-B). Solve volume word problems; recognise prisms with the same volume can have different dimensions (3DS-B).	Extended -- Stage 4 Area, Volume & Indices Apply $A = bh$ (parallelogram) and $A = \frac{1}{2}bh$ (triangle); find areas of composite figures (MA4-ARE-C-01). Use $V = \text{base area} \times \text{height}$ ($V = Ah$) for prisms with a uniform cross-section, including triangular prisms (MA4-VOL-C-01). Apply divisibility tests; write a whole number as a product of its prime factors in index notation (MA4-IND-C-01).

Teaching model

Students work independently through the Word Labs SOLO Tracker (wordlabs.app/solo) across differentiated bands -- Grow (resources), Know (practice), Show (auto-marked assessments). The teacher monitors live dashboard data and pulls small groups for targeted mini masterclasses (15-40 min) based on which outcomes have the most incomplete Show results. Remaining session time is used for incidental one-on-one and small group teaching in response to student questions and tracker data.

Working Mathematically (MAO-WM-01)

Process	How it appears in this unit
Communicating	Students use the terms product, factor, prime, area and volume, and record methods for finding areas and volumes in words.
Understanding and fluency	Students multiply whole numbers, calculate areas of rectangles, parallelograms and triangles, and volumes of rectangular prisms in cm ³ and m ³ .
Reasoning	Students reason about prime/composite numbers, why a triangle is half a parallelogram, and the multiplicative (layer) structure of volume.
Problem solving	Students solve area and volume problems, explore same-perimeter and same-volume relationships, and (Stage 4) apply area/volume formulas and prime factorisation.

Differentiation

Support	HPGE / Extension
Students in this group:	Students in this group:

Pre-test

Date	
Format	
Band place ments / key findings	

Post-test / unit evaluation

Date	
Key findings	
Students who need further consolidation	
Next steps	

Resources

See Appendix for Resources

Word Labs SOLO Tracker: wordlabs.app/solo | NSW DoE Stage 3 Year B Unit 37 | NESA Mathematics K-10 Syllabus (2022)

Unit 37 -- Outcome Teaching Record Find the outcome/s you taught, write your notes, sign and date. Order follows the outcome sequence, not your teaching calendar.

Co de	Outcome	Syllabus content point/s addressed	Mini masterclass teaching notes	Sign	Date
R1	Use the term product; multiply 2 or 3 numbers; list the factors of a number.	MR-A: use the term product; determine factors for a given whole number MA3-MR-01 Year A	Mini Lesson 1		
R2	Determine whether a number is prime, composite or neither.	MR-A: determine whether a number is prime, composite or neither (0 or 1) MA3-MR-01 Year A	Mini Lesson 2		
R3	Use the area model to multiply a 2-digit number by a 1-digit number.	MR-A: use partitioning and place value (area model) to multiply 2-digit by 1-digit numbers MA3-MR-01 Year A	Mini Lesson 3		
R4	Calculate the area of a rectangle in square metres (and cm2, km2).	2DS-A: calculate areas of rectangles in square centimetres (cm2), square metres (m2) and square kilometres (km2) MA3-2DS-02 Year A	Mini Lesson 5		
R5	Show rectangles with the same perimeter can have different dimensions and areas.	GM-A / 2DS-A: recognise rectangles with the same perimeter may have different dimensions; compare their areas MA3-GM-02, MA3-2DS-02 Year A	Mini Lesson 6		
R6	Find the volume of a rectangular prism (layers) and calculate it in cm3.	3DS-B: cubes in one layer x number of layers; record the method; calculate volume in cm3 MA3-3DS-02 Year B	Mini Lesson 9		
Y1	Transform a parallelogram into a rectangle and calculate its area.	2DS-B: transform a parallelogram into a rectangle to find its area; record the method MA3-2DS-02, MA3-2DS-03 Year B	Mini Lesson 7		
Y2	Recognise a triangle as half a parallelogram and calculate its area.	2DS-B: compare a triangle to a parallelogram with the same base and height; record a method for the area of a triangle MA3-2DS-02 Year B	Mini Lesson 8		
Y3	Explain a cubic metre and calculate volumes of rectangular prisms in m3.	3DS-B: estimate and measure volumes in cubic metres; calculate volumes of rectangular prisms in m3 MA3-3DS-02 Year B	Mini Lesson 10		
Y4	Select efficient strategies to multiply up to 4-digit by 1- and 2-digit numbers.	MR-B: select and use efficient strategies to multiply whole numbers of up to 4 digits by 1- and 2-digit numbers MA3-MR-01 Year B	Mini Lesson 4		
Y5	Solve volume word problems; same volume can have different dimensions.	3DS-B: calculate volumes of rectangular prisms to solve problems; recognise prisms with the same volume may have different dimensions MA3-3DS-02 Year B	Mini Lesson 11		
G1	Apply area formulas for triangles & parallelograms and find composite areas.	Stage 4 ARE: $A = \frac{1}{2}bh$ (triangle), $A = bh$ (parallelogram); area of composite figures MA4-ARE-C-01 Stage 4 (Core)	Mini Lesson 12		

Co de	Outcome	Syllabus content point/s addressed	Mini masterclass teaching notes	Sign	Date
G2	Use $V = \text{base area} \times \text{height}$ ($V = Ah$) for prisms, including triangular prisms.	Stage 4 VOL: uniform cross-section = base area; $V = \text{base area} \times \text{height} = Ah$ MA4-VOL-C-01 Stage 4 (Core)	Mini Lesson 13		
G3	Apply divisibility tests; write a number as a product of its prime factors (index notation).	Stage 4 IND: divisibility tests for 2,3,4,5,6,10; product of prime factors using index notation MA4-IND-C-01 Stage 4 (Core)	Mini Lesson 14		

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Unit 37 -- Mini Lesson Sequence

Multiplicative Relations & Measurement | Stage 3 Year B

Supplement to the SOLO Teacher Program (pages 1-2). Print pages 1-2 only for annotation. Reference these lesson plans when planning mini masterclasses or when a reviewer requires evidence of planned lesson content.

Red band -- Foundation -- Mini Lessons 1-9

Yellow band -- On-level -- Mini Lessons 4-11

Green band -- Extension -- Mini Lessons 12-14

DoE lesson links: Lessons marked DoE Lesson X draw directly from NSW DoE Mathematics Stage 3 Unit 37. Steps have been condensed to 20-40 minutes for mini masterclass delivery. Resources referenced by their original resource numbers from the DoE unit.

Mini Lesson 1 Products and factors

R1 | Red band | DoE Lesson 1 (adapted)

Duration: 25-30 min

Outcomes: R1

Syllabus: MA3-MR-01, MAO-WM-01 | MR-A: use the term product; determine factors of a number

Linked to: DoE Lesson 1 (adapted) -- Lesson 1 -- Products and factors | Key resources: Resource 1 - sorted numbers, counters, arrays

Learning intention

Students are learning to:
use the term product and find all the factors of a whole number

Success criteria

Students can:

- * I can multiply 2 or 3 numbers and call the answer the product.
- * I can list all the factors of a number using factor pairs.
- * I can check a factor by dividing with no remainder.

Mini masterclass structure (25-30 min)

Activate (4 min): Write $4 \times 3 = 12$. Name 12 the product and 4 and 3 the factors. Ask for another way to make 12.

Model (10 min): Build arrays for 12: 1x12, 2x6, 3x4. List the factor pairs to get all factors of 12: 1,2,3,4,6,12. Show a factor divides with no remainder.

Guided practice (10 min): Students find all factors of 18, 24 and 36 using factor pairs and record the product of each pair.

Check (4 min): Exit question: list all the factors of 20. What is the product of 5 and 4?

Key vocabulary

product, factor, factor pair, multiply, divide, array, whole number

Differentiation

Support: Provide counters to build arrays; start with smaller numbers and find pairs together.
Extension: Find numbers with exactly two factors, and numbers with many factors (e.g. 36, 48).

Assessment

Students list factors and name products — links to R1 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 2 Prime, composite or neither

R2 | Red band | DoE Lesson 1 (adapted)

Duration: 25 min | **Outcomes:** R2 | Syllabus: MA3-MR-01, MAO-WM-01 | MR-A: prime, composite or neither

Linked to: DoE Lesson 1 (adapted) -- Lesson 1 -- Primes | Key resources: Resource 2 - primes, hundred chart, counters

Learning intention		Success criteria
Students are learning to: decide whether a number is prime, composite or neither and explain why		Students can: * I know a prime number has exactly two factors: 1 and itself. * I know a composite number has more than two factors. * I can explain why 0 and 1 are neither prime nor composite.
Mini masterclass structure (25 min)		
Activate (3 min): Ask: how many factors does 7 have? How many does 8 have? Introduce the words prime and composite. Model (10 min): Use arrays: 7 makes only a 1x7 array (prime); 8 makes 1x8 and 2x4 (composite). Show 1 has one factor and 0 cannot be a factor product, so both are neither. Guided practice (8 min): Students sort numbers 2-20 into prime, composite or neither and justify each choice by listing factors. Check (4 min): Exit question: is 15 prime or composite? Explain using its factors.		
Key vocabulary		Differentiation
prime, composite, neither, factor, exactly two factors, justify		Support: Give a factor list for each number so students count factors before deciding. Extension: Use a hundred chart to mark primes (Sieve of Eratosthenes) and look for patterns.
Assessment		Teaching notes (annotate here)
Students classify numbers as prime/composite/neither — links to R2 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:

Mini Lesson 3 The area model for multiplication [HANDS-ON]

R3 | Red band | DoE Lesson 1 (adapted)

Duration: 25-30 min

Outcomes: R3

Syllabus: MA3-MR-01, MAO-WM-01 | MR-A: partitioning, place value and the area model

Linked to: DoE Lesson 1 (adapted) -- Lesson 1 -- Area model multiplication | Key resources: Grid paper, partitioned rectangles, place value cards

Physical materials needed: grid paper, place value cards

Learning intention

Students are learning to:
use the area model to multiply a 2-digit number by a 1-digit number

Success criteria

Students can:

- * I can split a 2-digit number into tens and ones.
- * I can draw an area model and multiply each part.
- * I can add the parts to get the total product.

Mini masterclass structure (25-30 min)

Activate (4 min): Show 23×4 . Ask students to split 23 into $20 + 3$ before multiplying.

Model (10 min): Draw a rectangle split into 20 and 3. Multiply $20 \times 4 = 80$ and $3 \times 4 = 12$, then add: $80 + 12 = 92$. Connect to place value and partitioning.

Guided practice (10 min): Students use the area model for 34×3 , 26×5 and 47×6 on grid paper.

Check (4 min): Exit question: use the area model to work out 38×4 .

Key vocabulary

area model, partition, tens, ones, place value, product, multiply, add

Differentiation

Support: Pre-draw the partitioned rectangle and label the tens and ones for students.
Extension: Extend the area model to 3-digit by 1-digit, and 2-digit by 2-digit multiplication.

Assessment

Students multiply using the area model — links to R3 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 4 Efficient multi-digit multiplication

Y4 | Yellow band | DoE Lesson 8 (adapted)

Duration: 30 min | **Outcomes:** Y4 | Syllabus: MA3-MR-01, MAO-WM-01 | MR-B: multiply up to 4 digits by 1- and 2-digit numbers

Linked to: DoE Lesson 8 (adapted) -- Lesson 8 -- Estimating and calculating (efficient strategies) | Key resources: Resource 20 - better deal, place value chart

Learning intention		Success criteria
Students are learning to: choose and use an efficient strategy to multiply a number up to 4 digits by a 1- or 2-digit number		Students can: * I can multiply a 3- or 4-digit number by a 1-digit number. * I can multiply a number by a 2-digit number using place value. * I can estimate first to check my answer is reasonable.
Mini masterclass structure (30 min)		
Activate (4 min): Show 1234×6. Ask students to estimate ($1200 \times 6 = 7200$) before calculating. Model (12 min): Model the standard algorithm / extended area model for 1234×6, then 246×23 (split 23 into $20 + 3$). Compare strategies for efficiency. Guided practice (10 min): Students multiply 4-digit by 1-digit and 3-digit by 2-digit numbers, estimating first. Check (4 min): Exit question: estimate then calculate 327×14.		
Key vocabulary		Differentiation
multiply, product, estimate, place value, partition, efficient strategy, algorithm		Support: Allow the area model so each partial product is visible; provide a multiplication chart. Extension: Multiply 4-digit by 2-digit numbers and solve multi-step money problems.
Assessment		Teaching notes (annotate here)
Students multiply large numbers efficiently — links to Y4 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:

Mini Lesson 5 Area of rectangles in metric units [HANDS-ON]

R4 | Red band | DoE Lesson 2 (adapted)

Duration: 25-30 min

Outcomes: R4

Syllabus: MA3-2DS-02, MAO-WM-01 | 2DS-A: area of rectangles in cm², m², km²

Linked to: DoE Lesson 2 (adapted) -- Lesson 2 -- Area of rectangles in square metres | Key resources: Resource 3 - fence it, Resource 4 - fencing dimensions, metre ruler

Physical materials needed: metre ruler, grid paper, tape measure

Learning intention

Students are learning to:
calculate the area of a rectangle in square metres, and choose the right metric unit

Success criteria

Students can:

- * I can calculate area using length x width.
- * I can calculate areas of rectangles in square metres (m²).
- * I can choose cm², m² or km² to suit the size of the rectangle.

Mini masterclass structure (25-30 min)

Activate (4 min): Mark a 1 m x 1 m square on the floor. Ask what one square metre means.

Model (10 min): Calculate the area of a 4 m x 3 m classroom rug: $4 \times 3 = 12$ m². Discuss when to use cm² (a book cover), m² (a room) and km² (a suburb).

Guided practice (10 min): Students calculate areas of rectangles in m² and choose an appropriate unit for given objects.

Check (4 min): Exit question: a garden bed is 5 m by 2 m. What is its area? What unit did you use?

Key vocabulary

area, length, width, square metre, m², cm², km², unit, rectangle

Differentiation

Support: Provide grid overlays so students can count square metres before using length x width.
Extension: Convert between cm² and m², and calculate the area of an L-shaped region.

Assessment

Students calculate rectangle areas in metric units — links to R4 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 6 Same perimeter, different dimensions [HANDS-ON]

R5 | Red band | DoE Lesson 2 (adapted)

Duration: 25-30 min

Outcomes: R5

Syllabus: MA3-GM-02, MA3-2DS-02, MAO-WM-01 | GM-A / 2DS-A: same perimeter, different area

Linked to: DoE Lesson 2 (adapted) -- Lesson 2 -- Rectangles with the same perimeter | Key resources: Resource 4 - fencing dimensions, grid paper, string

Physical materials needed: grid paper, string / loop

Learning intention

Students are learning to:
show that rectangles with the same perimeter can have different dimensions and areas

Success criteria

Students can:

- * I can draw different rectangles with the same perimeter.
- * I can calculate and compare their areas.
- * I can explain that the same perimeter does not mean the same area.

Mini masterclass structure (25-30 min)

Activate (4 min): Give students a loop of string. Ask them to make two different rectangles with the same perimeter.

Model (10 min): Using a perimeter of 24 m: draw 1x11, 4x8 and 6x6. Calculate areas (11, 32, 36 m²). Show the square gives the largest area.

Guided practice (10 min): Students list rectangles with a perimeter of 20 m and compare their areas on grid paper.

Check (4 min): Exit question: two rectangles both have a perimeter of 16 m. Can they have different areas? Show an example.

Key vocabulary

perimeter, dimensions, area, length, width, rectangle, compare, square

Differentiation

Support: Provide grid paper with the perimeter pre-counted so students focus on area.
Extension: Investigate which rectangle of a fixed perimeter has the maximum area and why.

Assessment

Students compare same-perimeter rectangles — links to R5 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 7 Area of a parallelogram [HANDS-ON]

Y1 | Yellow band | DoE Lesson 3 (adapted)

Duration: 30 min | **Outcomes:** Y1 | Syllabus: MA3-2DS-02, MA3-2DS-03, MAO-WM-01 | 2DS-B: transform a parallelogram into a rectangle

Linked to: DoE Lesson 3 (adapted) -- Lesson 3 -- Calculating the area of a parallelogram | Key resources: Resource 6 - parallelogram proof, paper parallelograms, scissors

Physical materials needed: paper parallelograms, scissors, ruler

Learning intention	Success criteria
Students are learning to: transform a parallelogram into a rectangle and calculate its area	Students can: * I can cut a right-angled piece off a parallelogram and move it to make a rectangle. * I can record a method for finding the area of a parallelogram. * I can calculate the area using base x perpendicular height.

Mini masterclass structure (30 min)

Activate (4 min): Show a parallelogram. Ask: how could we turn this into a rectangle we already know how to measure?

Model (12 min): Cut a right-angled triangle off one end of a paper parallelogram and slide it to the other end to form a rectangle. The base and height stay the same, so area = base x height.

Guided practice (10 min): Students cut and rearrange parallelograms, then calculate areas using base x perpendicular height.

Check (4 min): Exit question: a parallelogram has base 8 cm and height 5 cm. What is its area?

Key vocabulary	Differentiation
parallelogram, base, perpendicular height, rectangle, dissect, rearrange, area	Support: Pre-cut the parallelogram along the height line so students focus on rearranging. Extension: Find the area when only a slant side is given and explain why height (not slant) is used.
Assessment	Teaching notes (annotate here)
Students find the area of a parallelogram — links to Y1 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 8 Area of a triangle [HANDS-ON]

Y2 | Yellow band | DoE Lesson 4 (adapted)

Duration: 30 min | **Outcomes:** Y2 | Syllabus: MA3-2DS-02, MAO-WM-01 | 2DS-B: a triangle is half a parallelogram

Linked to: DoE Lesson 4 (adapted) -- Lesson 4 -- Calculating the area of a triangle | Key resources: Resource 8 - triangle area, paper triangles, scissors

Physical materials needed: paper triangles, scissors, ruler

Learning intention		Success criteria	
Students are learning to: recognise a triangle as half a parallelogram and calculate its area		Students can: * I can join two identical triangles to make a parallelogram. * I know a triangle has half the area of a parallelogram with the same base and height. * I can calculate the area of a triangle using $\frac{1}{2} \times \text{base} \times \text{height}$.	
Mini masterclass structure (30 min)			
Activate (4 min): Give pairs two identical triangles. Ask them to make a parallelogram from them.			
Model (12 min): Show two copies of a triangle form a parallelogram, so the triangle is half. Area of triangle = $\frac{1}{2} \times \text{base} \times \text{height}$. Work an example: base 10 cm, height 6 cm -> $\frac{1}{2} \times 10 \times 6 = 30 \text{ cm}^2$.			
Guided practice (10 min): Students calculate areas of triangles using $\frac{1}{2} \times \text{base} \times \text{height}$, identifying the perpendicular height.			
Check (4 min): Exit question: a triangle has base 12 cm and height 5 cm. What is its area?			
Key vocabulary		Differentiation	
triangle, parallelogram, base, perpendicular height, half, area, $\frac{1}{2} \times \text{base} \times \text{height}$		Support: Provide pre-cut identical triangles so students can build the parallelogram first. Extension: Find the area of triangles where the height is outside the triangle (obtuse triangles).	
Assessment		Teaching notes (annotate here)	
Students find the area of a triangle — links to Y2 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:	

Mini Lesson 9 Volume of rectangular prisms (cubic centimetres) [HANDS-ON]

R6 | Red band | DoE Lessons 5 & 6 (adapted)

Duration: 30 min | **Outcomes:** R6 | Syllabus: MA3-3DS-02, MAO-WM-01 | 3DS-B: layer structure of volume; volume in cm³

Linked to: DoE Lessons 5 & 6 (adapted) -- Lessons 5 & 6 -- Volume of rectangular prisms | Key resources: Resource 9 - box of boxes, Resource 15 - calculate the volume, centicubes

Physical materials needed: centicubes, rectangular boxes

Learning intention	Success criteria
Students are learning to: find the volume of a rectangular prism by layers and calculate it in cubic centimetres	Students can: * I can find the number of cubes in one layer. * I can multiply the cubes in one layer by the number of layers. * I can calculate the volume of a rectangular prism in cubic centimetres (cm³).

Mini masterclass structure (30 min)

Activate (4 min): Build one layer of centicubes (e.g. $4 \times 3 = 12$). Ask how many cubes if we stack 5 layers.

Model (12 min): Show volume = cubes in one layer $\times$ number of layers = $12 \times 5 = 60$ cm³. Connect to length $\times$ width $\times$ height ($4 \times 3 \times 5$). Record the method in words.

Guided practice (10 min): Students calculate volumes of rectangular prisms in cm³ using the layer method and $L \times W \times H$.

Check (4 min): Exit question: a prism is 5 cm $\times$ 2 cm $\times$ 4 cm. What is its volume?

Key vocabulary	Differentiation
volume, rectangular prism, layer, cubic centimetre, cm ³ , length, width, height	Support: Build the prism with centicubes and count one layer before multiplying. Extension: Find a missing dimension when the volume and two other dimensions are given.
Assessment	Teaching notes (annotate here)
Students calculate volume in cm ³ — links to R6 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 10 Volume in cubic metres [HANDS-ON]

Y3 | Yellow band | DoE Lesson 7 (adapted)

Duration: 25-30 min

Outcomes: Y3

Syllabus: MA3-3DS-02, MAO-WM-01 | 3DS-B: cubic metres; volume of prisms in m3

Linked to: DoE Lesson 7 (adapted) -- Lesson 7 -- Solving problems involving volume (cubic metres) | Key resources: Resource 16 - Lisa's landscaping, Resource 17 - estimating volume, metre rulers

Physical materials needed: metre rulers, tape measure

Learning intention

Students are learning to:
explain a cubic metre and calculate volumes of rectangular prisms in cubic metres

Success criteria

Students can:

- * I can explain a cubic metre (1 m x 1 m x 1 m) as a measure of volume.
- * I can calculate volumes of rectangular prisms in cubic metres (m3).
- * I can choose cm3 or m3 to suit the size of the object.

Mini masterclass structure (25-30 min)

Activate (4 min): Use metre rulers to outline a cubic metre in a corner of the room. Ask what could fit inside it.

Model (10 min): Calculate the volume of a 3 m x 2 m x 2 m shipping container: $3 \times 2 \times 2 = 12 \text{ m}^3$. Discuss when m3 is sensible versus cm3.

Guided practice (10 min): Students calculate volumes of prisms in m3 (soil, water tanks, rooms) and choose appropriate units.

Check (4 min): Exit question: a sandpit is 2 m x 2 m x 1 m. What is its volume in m3?

Key vocabulary

volume, cubic metre, m3, cubic centimetre, cm3, rectangular prism, estimate, unit

Differentiation

Support: Anchor the size of a cubic metre with the metre-ruler model before calculating.
Extension: Estimate the volume of large real spaces (e.g. the classroom) and compare with a calculation.

Assessment

Students calculate volume in m3 — links to Y3 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 11 Solving problems involving volume **[HANDS-ON]**

Y5 | Yellow band | DoE Lessons 7 & 8 (adapted)

Duration: 30 min **Outcomes:** Y5 Syllabus: MA3-3DS-02, MAO-WM-01 | 3DS-B: volume problems; same volume, different dimensions

Linked to: DoE Lessons 7 & 8 (adapted) -- Lessons 7 & 8 -- Solving problems involving volume | Key resources: Resource 10 - recording boxes, Resource 11 - bigger cubes, centicubes

Physical materials needed: centicubes

Learning intention	Success criteria
Students are learning to: solve volume word problems and show prisms with the same volume can have different dimensions	Students can: * I can solve a word problem by finding the volume of a rectangular prism. * I can find different prisms that have the same volume. * I can find a missing dimension when I know the volume.

Mini masterclass structure (30 min)

Activate (4 min): Ask: how many different boxes can hold exactly 24 cubes? Predict before building.

Model (12 min): Find prisms with volume 24 cm³: 1x1x24, 2x3x4, 2x2x6. Then solve a word problem: a tank holds 36 m³; if it is 3 m x 3 m, how deep is it? ($36 \div 9 = 4$ m).

Guided practice (10 min): Students solve volume word problems and list prisms with a given volume.

Check (4 min): Exit question: a box has a volume of 48 cm³ and a base of 4 cm x 3 cm. How tall is it?

Key vocabulary	Differentiation
volume, rectangular prism, dimensions, missing dimension, divide, word problem	Support: Provide centicubes to build prisms for the given volume before reasoning abstractly. Extension: Find every whole-number prism with a given volume and describe the patterns.
Assessment	Teaching notes (annotate here)
Students solve volume problems — links to Y5 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 12 Area formulas and composite figures

G1 | Green band | Original mini lesson (Stage 4 extension)

Duration: 30 min | **Outcomes:** G1 | Syllabus: MA4-ARE-C-01, MAO-WM-01 | Stage 4: $A = bh$, $A = \frac{1}{2}bh$, composite areas

Learning intention		Success criteria	
Students are learning to: apply the area formulas for triangles and parallelograms and find the area of composite figures		Students can: * I can use $A = bh$ to find the area of a parallelogram. * I can use $A = \frac{1}{2}bh$ to find the area of a triangle. * I can split a composite shape into rectangles and triangles and add the areas.	
Mini masterclass structure (30 min)			
Activate (4 min): Review base x height for a parallelogram and $\frac{1}{2}$ x base x height for a triangle. Model (12 min): Apply the formulas to several shapes, then split an L-shape / arrow into a rectangle and a triangle, find each area and add them. Guided practice (10 min): Students calculate composite areas by dissecting shapes into rectangles, parallelograms and triangles. Check (4 min): Exit question: find the area of a shape made of a 6x4 rectangle with a triangle (base 6, height 3) on top.			
Key vocabulary		Differentiation	
formula, base, perpendicular height, parallelogram, triangle, composite figure, dissect		Support: Provide the shape pre-split with dashed lines so students apply each formula in turn. Extension: Find composite areas that include a missing length students must work out first.	
Assessment		Teaching notes (annotate here)	
Students apply area formulas and find composite areas — extends Y1/Y2.		Taught as described / adapted / replaced -- note here:	

Mini Lesson 13 Volume of prisms ($V = \text{base area} \times \text{height}$)

G2 | Green band | Original mini lesson (Stage 4 extension)

Duration: 30 min | **Outcomes:** G2 | Syllabus: MA4-VOL-C-01, MAO-WM-01 | Stage 4: $V = Ah$ for prisms with uniform cross-section

Learning intention		Success criteria
Students are learning to: use $V = \text{base area} \times \text{height}$ to find the volume of prisms, including triangular prisms		Students can: * I can identify the uniform cross-section of a prism and find its area. * I can use $V = \text{base area} \times \text{height}$ ($V = Ah$). * I can find the volume of a triangular prism.
Mini masterclass structure (30 min)		
Activate (4 min): Show that a rectangular prism is base area $\times$ height. Ask what the base area is for a triangular prism. Model (12 min): Establish $V = Ah$. For a triangular prism with a triangle base (base 4, height 3 $\rightarrow A = 6$) and length 10: $V = 6 \times 10 = 60 \text{ cm}^3$. Guided practice (10 min): Students find the volume of rectangular and triangular prisms using $V = \text{base area} \times \text{height}$. Check (4 min): Exit question: a triangular prism has a base area of 9 cm^2 and length 7 cm. What is its volume?		
Key vocabulary		Differentiation
prism, uniform cross-section, base area, height, length, $V = Ah$, triangular prism		Support: Provide the base area so students focus on multiplying by the length. Extension: Find the volume of prisms with trapezium or compound cross-sections.
Assessment		Teaching notes (annotate here)
Students apply $V = Ah$ to prisms — extends R6/Y3/Y5.		Taught as described / adapted / replaced -- note here:

Mini Lesson 14 Prime factorisation and divisibility tests G3 Green band Original mini lesson (Stage 4 extension)	
Duration: 30 min	Outcomes: G3 Syllabus: MA4-IND-C-01, MAO-WM-01 Stage 4: divisibility tests; product of primes in index notation
Learning intention	Success criteria
Students are learning to: apply divisibility tests and write a whole number as a product of its prime factors using index notation	Students can: * I can use divisibility tests for 2, 3, 4, 5, 6 and 10. * I can use a factor tree to find the prime factors of a number. * I can write the product of prime factors using index notation (e.g. $72 = 2^3 \times 3^2$).
Mini masterclass structure (30 min)	
Activate (4 min): Check divisibility: is 84 divisible by 2, 3 and 6? Use the digit-sum test for 3. Model (12 min): Build a factor tree for 72: $72 = 8 \times 9 = 2 \times 2 \times 2 \times 3 \times 3$. Write $72 = 2^3 \times 3^2$ using index notation. Guided practice (10 min): Students make factor trees for 36, 60 and 48 and write each as a product of primes in index form. Check (4 min): Exit question: write 60 as a product of its prime factors using index notation.	
Key vocabulary	Differentiation
prime factor, factor tree, product of primes, index notation, power, divisibility test	Support: Provide the first split of the factor tree so students continue to the primes. Extension: Use prime factorisation to find the highest common factor of two numbers.
Assessment	Teaching notes (annotate here)
Students write numbers as products of primes — extends R1/R2.	Taught as described / adapted / replaced -- note here:

Unit 37 -- Resource Appendix

Every link verified. Videos open in a browser; worksheets are print-ready PDFs (see the download checklist at the end).

Type	Resource	Link
R1 -- Use the term product; multiply 2 or 3 numbers; list the factors of a number.		
Video	Factors Common Factors Greatest Common Factor (GCF) Math with Mr. J	https://www.youtube.com/watch?v=IRHwKNBpG_Q
Video	Multiples vs. Factors What are Multiples and Factors? Math with Mr. J	https://www.youtube.com/watch?v=1qPflkOEuYY
Worksheet (print)	Corbettmaths — Factors (PDF)	https://corbettmaths.com/wp-content/uploads/2018/11/Factors-216-pdf.pdf
R2 -- Determine whether a number is prime, composite or neither.		
Video	Prime and Composite Numbers Math with Mr. J	https://www.youtube.com/watch?v=ydm2cxacPIM
Video	What are Prime Numbers? Math with Mr. J	https://www.youtube.com/watch?v=5gBtzdM5M5k
Worksheet (print)	Corbettmaths — Prime Numbers (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/prime-numbers-pdf.pdf
R3 -- Use the area model to multiply a 2-digit number by a 1-digit number.		
Video	Box Method Multiplication 2-Digits x 2-Digits Math with Mr. J	https://www.youtube.com/watch?v=n3q3XzzIGSY
Video	Multiplication: A Step-By-Step Review How to Multiply Math with Mr. J	https://www.youtube.com/watch?v=g5TmDrXeIR8
Worksheet (print)	Corbettmaths — Multiplication (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/multiplication-pdf.pdf
R4 -- Calculate the area of a rectangle in square metres (and cm², km²).		
Video	How to Find the Area of Rectangles and Squares Math with Mr. J	https://www.youtube.com/watch?v=_ejDO6aPPzU
Video	How to Find the Area of a Rectangle Math with Mr. J	https://www.youtube.com/watch?v=vqJmXGIF_1w
Worksheet (print)	Corbettmaths — Area of a Rectangle (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/area-of-a-rectangle-pdf.pdf
R5 -- Show rectangles with the same perimeter can have different dimensions and areas.		
Video	Same Perimeter — Different Areas (Grade 3) MATH-N-ROLL	https://www.youtube.com/watch?v=0Ho8Az8y4Mw
Video	Same Perimeter — Different Area (Grade 3 Math) JoAnn's School	https://www.youtube.com/watch?v=R4SNnVceMz4
Worksheet (print)	Corbettmaths — Area of Rectangles (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/area-of-rectangles-pdf.pdf
R6 -- Find the volume of a rectangular prism (layers) and calculate it in cm³.		
Video	Finding Volume with Unit Cubes How to Find Volume Math with Mr. J	https://www.youtube.com/watch?v=RxkRIIAucMk
Video	Volume of Rectangular Prisms Math with Mr. J	https://www.youtube.com/watch?v=By7sVb2lhFs
Worksheet (print)	Corbettmaths — Volume of a Cuboid (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/volume-of-a-cuboid-pdf.pdf
Y1 -- Transform a parallelogram into a rectangle and calculate its area.		
Video	Area of Parallelograms How to Find the Area of a Parallelogram Math with Mr. J	https://www.youtube.com/watch?v=uj6k22WubCk
Video	Area of a Parallelogram (6th Grade Math) JoAnn's School	https://www.youtube.com/watch?v=pNnZ-WgK0fs
Worksheet (print)	Corbettmaths — Area of a Parallelogram (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/area-of-a-parallelogram-pdf1.pdf
Y2 -- Recognise a triangle as half a parallelogram and calculate its area.		
Video	Finding the Area of a Triangle A Step-By-Step Guide Math with Mr. J	https://www.youtube.com/watch?v=-ZPFB1uDjEw
Video	How to Find the Area and Perimeter of a Triangle Math with Mr. J	https://www.youtube.com/watch?v=JCWJihpZ-Lo
Worksheet (print)	Corbettmaths — Area of a Triangle (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/area-of-a-triangle-pdf.pdf
Y3 -- Explain a cubic metre and calculate volumes of rectangular prisms in m³.		

Type	Resource	Link
Video	Volume of Rectangular Prisms Math with Mr. J	https://www.youtube.com/watch?v=By7sVb2lhFs
Video	Volume of a Rectangular Prism (Mastery Check) Math with Mr. J	https://www.youtube.com/watch?v=4i_XprDGeVs
Worksheet (print)	Corbettmaths — Volume of a Cuboid (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/volume-of-a-cuboid-pdf.pdf
Y4 -- Select efficient strategies to multiply up to 4-digit by 1- and 2-digit numbers.		
Video	Multiplication: A Step-By-Step Review How to Multiply Math with Mr. J	https://www.youtube.com/watch?v=g5TmDrXeIR8
Video	2-Digit Multiplication: A Step-By-Step Review Math with Mr. J	https://www.youtube.com/watch?v=yiZ3r9T_OJs
Worksheet (print)	Corbettmaths — Multiplication (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/multiplication-pdf.pdf
Y5 -- Solve volume word problems; same volume can have different dimensions.		
Video	Volume of Rectangular Prism Word Problems Wrestle with Math	https://www.youtube.com/watch?v=gIFR36JiKtI
Video	Volume of Composite Rectangular Prisms Math with Mr. J	https://www.youtube.com/watch?v=dAKAfcqj3x4
Worksheet (print)	Corbettmaths — Volume of a Cuboid (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/volume-of-a-cuboid-pdf.pdf
G1 -- Apply area formulas for triangles & parallelograms and find composite areas.		
Video	Finding the Area of Composite Figures with Triangles Math with Mr. J	https://www.youtube.com/watch?v=bo483Varm-U
Video	Finding the Area of a Composite Figure Area of Composite Rectangles Math with Mr. J	https://www.youtube.com/watch?v=z4Lat1uOQI4
Worksheet (print)	Corbettmaths — Area of Compound Shapes (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/area-of-compound-shapes-pdf.pdf
G2 -- Use $V = \text{base area} \times \text{height}$ ($V = Ah$) for prisms, including triangular prisms.		
Video	How to Find the Volume of a Triangular Prism Math with Mr. J	https://www.youtube.com/watch?v=4pV7_NuGO2g
Video	Volume of Composite Rectangular Prisms Math with Mr. J	https://www.youtube.com/watch?v=dAKAfcqj3x4
Worksheet (print)	Corbettmaths — Volume of a Prism (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/volume-of-a-prism-pdf.pdf
G3 -- Apply divisibility tests; write a number as a product of its prime factors (index notation).		
Video	Prime Factorization What is Prime Factorization? Math with Mr. J	https://www.youtube.com/watch?v=EXU4ai9aViE
Video	Prime Factorization of 72 Math with Mr. J	https://www.youtube.com/watch?v=kg4DP0QRL6c
Worksheet (print)	Corbettmaths — Product of Primes (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/product-of-primes-pdf1.pdf
Beyond SOLO -- Stage 4 extension		
Video	Finding the Area of Composite Figures with Triangles Math with Mr. J	https://www.youtube.com/watch?v=bo483Varm-U
Video	How to Find the Volume of a Triangular Prism Math with Mr. J	https://www.youtube.com/watch?v=4pV7_NuGO2g
Video	Prime Factorization of 72 Math with Mr. J	https://www.youtube.com/watch?v=kg4DP0QRL6c
Worksheet (print)	Corbettmaths — Volume of a Prism (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/volume-of-a-prism-pdf.pdf
Worksheet (print)	Corbettmaths — Area of Compound Shapes (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/area-of-compound-shapes-pdf.pdf

Worksheet download checklist

Download and print these PDFs before the unit (or save to your class drive). Tick each as you go.

- ☐ Corbettmaths — Factors (PDF) -- <https://corbettmaths.com/wp-content/uploads/2018/11/Factors-216-pdf.pdf>
- ☐ Corbettmaths — Prime Numbers (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/prime-numbers-pdf.pdf>
- ☐ Corbettmaths — Multiplication (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/multiplication-pdf.pdf>
- ☐ Corbettmaths — Area of a Rectangle (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/area-of-a-rectangle-pdf.pdf>
- ☐ Corbettmaths — Area of Rectangles (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/area-of-rectangles-pdf.pdf>
- ☐ Corbettmaths — Volume of a Cuboid (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/volume-of-a-cuboid-pdf.pdf>

- ☐ Corbettmaths — Area of a Parallelogram (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/area-of-a-parallelogram-pdf1.pdf>
- ☐ Corbettmaths — Area of a Triangle (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/area-of-a-triangle-pdf.pdf>
- ☐ Corbettmaths — Area of Compound Shapes (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/area-of-compound-shapes-pdf.pdf>
- ☐ Corbettmaths — Volume of a Prism (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/volume-of-a-prism-pdf.pdf>
- ☐ Corbettmaths — Product of Primes (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/product-of-primes-pdf1.pdf>